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Information extraction for materials science using large language models: A survey of recent advances and new frontiers

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ABSTRACT

Information Extraction (IE) transforms unstructured documents into structured information and is essential for accelerating data-driven materials discovery. However, materials science literature poses unique challenges for IE because it contains specialized terminology, ambiguous material expressions, complex experimental descriptions, and heterogeneous data formats. Recently, Large Language Models (LLMs) have shown strong potential for extracting and organizing material knowledge, but their effectiveness for domain-specific IE tasks and their ability to address persistent extraction challenges remain insufficiently synthesized. This survey provides a systematic review of LLM-based IE in materials science. Unlike broader reviews on LLMs for materials discovery, this survey focuses specifically on the information-extraction layer that converts unstructured materials literature into structured and reusable knowledge. We introduce a taxonomy of key IE tasks, ranging from Named Entity Recognition (NER) and Relation Extraction (RE) to multi-modal extraction, classification, and generative data structuring. We then review the major methodological paradigms, including prompting, finetuning, retrieval-augmented generation, and ensemble or agent-based methods, together with domain-specific datasets, pretrained and foundation models, tools, and evaluation protocols. Overall, this survey organizes the field into six task categories, four methodological paradigms, three resource groups, three representative application scenarios, and six major challenge areas. Finally, we identify six main challenges for future research: (1) inconsistent material terminology, (2) uneven domain coverage across materials subfields, (3) complex multi-modal data integration, (4) domain-specific data scarcity, (5) practical deployment issues of LLM-based IE, and (6) the need for rigorous, fact-checking evaluation protocols. This work provides a structured guide for developing reliable LLM-based IE pipelines for materials science.

1. Introduction

Recent advances in experimental and computational materials science have substantially increased the volume of scientific publications [1,2]. This growth is further accelerated by the expanding use of machine learning, physics-informed modeling, and data-driven design across materials and engineering problems, including alloy design, property prediction, structural response analysis, and biomaterials research [3–7]. These publications contain rich knowledge about material compositions, synthesis procedures, and properties, which are essential for the discovery of data-driven materials [8,9]. However, much of this knowledge is reported in unstructured formats, such as narrative text, complex tables, and images, rather than in standardized and machine-readable forms [10,11]. This issue is particularly evident in image-rich materials and biomaterials studies, where microscopy and bioimage data often contain key structural and functional information that require automated analysis [12]. Unlike structured databases, these formats lack a fixed organization that computers can directly process; as a result, critical information cannot be easily extracted, compared, or reused in data-driven research pipelines [13,14]. This creates a major bottleneck: key scientific details remain inaccessible within the

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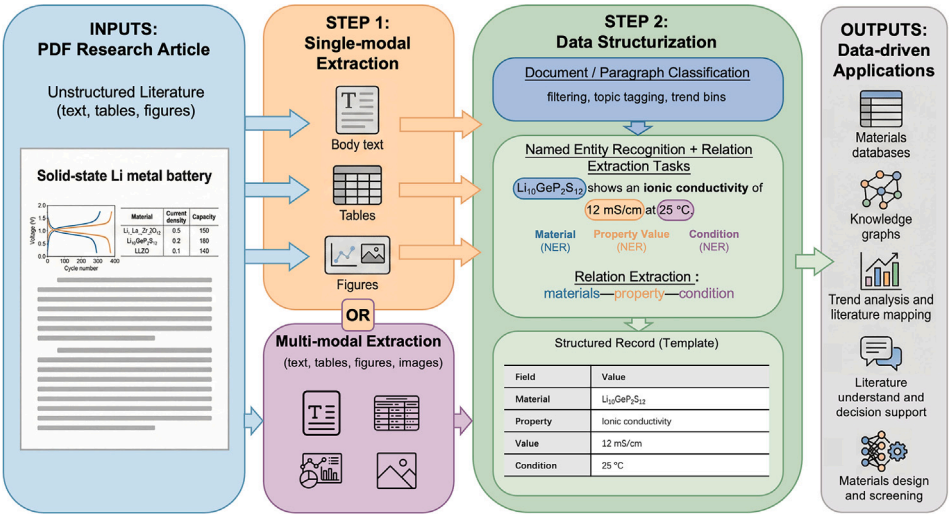


Fig. 1. Overview of the information extraction (IE) workflow for transforming PDF materials articles into structured records. The pipeline connects document parsing and core extraction tasks to downstream applications such as database construction and decision support.

unstructured literature and cannot be leveraged for large-scale analysis without systematic extraction [15,16].

Information Extraction (IE) directly addresses this bottleneck by automatically transforming unstructured information into structured datasets [17,18]. In fast-growing domains such as batteries, catalysts, polymers, and alloys [19–21], IE enables more efficient aggregation of materials data and substantially reduces duplicate manual effort during literature review and curation [22,23]. Recent examples further show that data-driven strategies are being applied to increasingly diverse materials systems and characterization problems, including self-healing microporous materials, high-entropy alloys, and hyperspectral characterization of iron tailings for resource evaluation and reuse [24–26]. The resulting structured resources serve as the foundation for materials databases, large-scale trend analysis, and the development of data-driven models [27–29]. Furthermore, IE plays a pivotal role in advanced computational workflows: when integrated with machine learning methods, extracted data can inform material design [30], narrow promising parameter spaces [31], and support automated hypothesis generation and knowledge discovery [23,32]. Consequently, IE has become a fundamental enabling component for scalable, data-centric research in modern materials science. To make this process more concrete, Fig. 1 illustrates the type of information extraction workflow considered in this survey. A materials research article in PDF format is first parsed into separate segments, including body text, tables, and figures. IE components such as named entity recognition (NER), relation extraction (RE) and template filling then convert sentences, such as those reporting the ionic conductivity of $Li_{10}GeP_2S_{12}$ at 25 °C, into structured records. The resulting structured data provide a common starting point for building materials databases, knowledge graphs, trend analyses, and other applications that we discuss in later Sections 5.

Despite the conceptual clarity of the IE workflow outlined above, its practical realization has historically relied on rule-based methods or conventional deep learning models, such as convolutional or recurrent neural networks [13,15,33]. These early approaches achieved foundational success, for instance, in identifying simple chemical formulas or compound names from materials science papers [30]. However, they struggled to cope with the linguistic complexity of scientific writing, which includes specialized terminology, semantic ambiguity, and long-range contextual dependencies [34,35]. As a result, traditional models often failed to recognize that abbreviations such as “LCO” and their full forms (e.g., “lithium cobalt oxide”) refer to the same material, or to disambiguate whether expressions like “high yield” describe a material property or a synthesis quantity. Moreover, these models frequently could not reliably associate a reported property value (e.g., “150 mAh/g”) with the correct material when the two were separated by multiple sentences or paragraph boundaries. These challenges highlight a fundamental gap in earlier IE systems: the lack of mechanisms for unified semantic representation and long-range contextual reasoning across scientific text.

To address these challenges, recent advances in Large Language Models (LLMs) have introduced a novel paradigm for information extraction in materials science [36]. Built on the Transformer [37] and pretrained on large-scale text corpora [38], LLMs enable context-aware semantic modeling through self-attention mechanisms, allowing them to capture long-range dependencies and resolve ambiguities that were difficult for earlier IE approaches. Depending on their architectural design, LLMs can be broadly categorized into encoder-only models such as BERT [39] and SciBERT [34], decoder-only models such as the GPT series [40] and LLaMA [41], and encoder–decoder models such as T5 [42] and BART [43], which provide complementary strengths in text understanding, prompt-based extraction, and text-to-structure transformation. These capabilities have motivated a growing body of work applying LLMs to materials science, including high-level explorations of their potential impact [44] and application-driven studies in materials discovery and sustainability [45]. However, existing reviews predominantly emphasize downstream applications or chemistry-centric extraction tasks [36,46], or discuss generic “generative IE” frameworks from a computer science perspective [47]. As a result, the field still lacks a task-level, materials-specific understanding of information extraction methods that account for heterogeneous data modalities and domain-dependent semantics.

To address this gap, this survey provides a systematic, task-oriented review of LLM-based information extraction in materials science. Rather than organizing the literature around downstream applications, we adopt a bottom-up perspective that starts from information extraction itself, examining how core IE techniques can be applied, evaluated, and extended across heterogeneous materials data. Specifically, we focus on the fundamental steps required to construct structured knowledge from the unstructured scientific literature, including the identification of key entities and relations, the extraction of information from text, tables, and figures, and the integration of extracted data into higher-level applications such as material databases, knowledge graphs, and trend analysis. Covering developments since 2023, when LLM-based methods began to be widely adopted in this domain [48], this survey

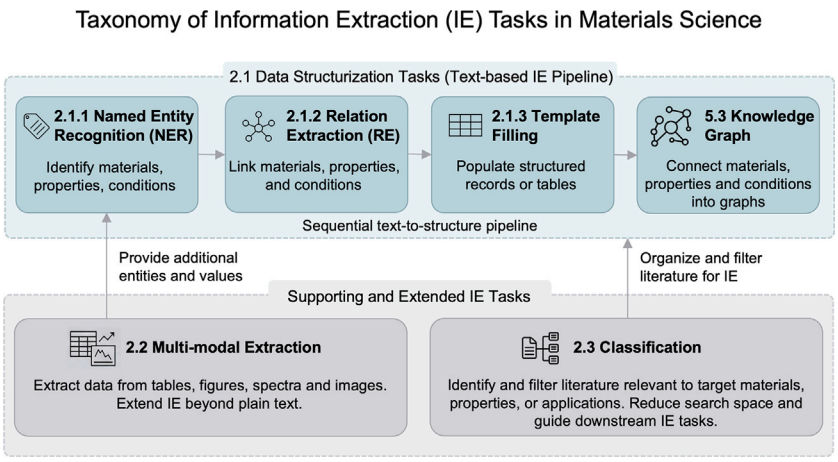


Fig. 2. Taxonomy of IE tasks in materials science, showing the core text-based data structuring pipeline and the supporting tasks of multi-modal extraction and classification.

aims to serve as a practical guide for AI researchers and materials scientists, helping to consolidate existing efforts, reduce duplicated work, and clarify the methodological foundations needed for reliable, data-centric materials discovery. The novelty of this survey lies in its task-level and resource-aware treatment of LLM-based IE for materials science. Unlike broader reviews that mainly discuss LLMs as general-purpose tools for materials discovery, this survey focuses specifically on the information-extraction layer that converts unstructured materials literature into structured and reusable knowledge. This perspective allows us to connect IE tasks, methodological choices, datasets, applications, and evaluation challenges within a single materials-oriented framework.

Overall, the main contribution of this survey can be summarized as follows.

- **Comprehensive Survey of LLM-based IE for Materials Science.** We provide a comprehensive overview of recent LLM-based information extraction research in materials science, with a particular focus on how LLMs are applied to domain-specific IE tasks and whether they can overcome persistent challenges in scientific materials texts.
- **Task-Level Taxonomy of Materials Science IE.** We introduce a task-level taxonomy that organizes key IE problems in materials science, ranging from core tasks such as named entity recognition and relation extraction to emerging directions including multi-modal extraction and generative data structuring, thereby providing a structured roadmap from methods to research objectives.
- **Methods, Resources, and Evaluation Protocols.** We systematically review the methodological shift toward prompting and fine-tuning, and summarize essential domain-specific resources and evaluation protocols that support reliable model development, benchmarking, and assessment across heterogeneous materials data.
- **Challenges and Future Directions.** We distill six persistent challenges that shape the future agenda of LLM-based IE in materials science, including (1) inconsistent material terminology, (2) uneven domain coverage across materials subfields, (3) complex multi-modal data integration, (4) domain-specific data scarcity, (5) the need for fact-checking evaluation protocols, and (6) practical deployment issues of LLM-based IE, and discuss the corresponding opportunities for future research.

To give a clear and systematic view of how these steps can be performed, this survey is organized as follows. Section 2 presents a task-level taxonomy of IE in materials science. Section 3 reviews the methods used to address these tasks, followed by Section 4, which summarizes key datasets and resources that enable model development and evaluation. Section 5 highlights important applications driven by IE, and Section 6 summarizes the main challenges and opportunities for future research in this area. Finally, Section 7 concludes the survey.

2. Taxonomy of tasks in materials science IE

Information extraction in materials science involves several interdependent tasks that collectively convert the scientific literature into machine-usable knowledge. These tasks differ in purpose, level of granularity, and output type, but together form the functional landscape of IE research in this field. In this survey, we categorize IE tasks into two major groups: (1) **data structurization tasks**, which transform unstructured text into structured representations, and (2) **supporting and extended IE tasks**, such as multi-modal extraction and document classification, which either complement structurization or enable downstream use cases. Within data structurization, we follow a pipeline view consisting of four sequential tasks: named entity recognition (NER), relation extraction (RE), template filling, and abstractive summarization. These should be interpreted as ordered steps rather than independent objectives. Conversely, multi-modal extraction addresses information beyond text, while classification organizes literature at the document or topic level for retrieval and large-scale analysis. Fig. 2 shows an overview of this taxonomy, with detailed discussions presented in the following subsections.

2.1. Data structurization

Data structurization transforms unstructured textual information into curated, machine-readable representations, for example through template filling and structured summarization [49]. This process is critical in scientific literature, where materials-related knowledge is often expressed in heterogeneous textual forms and dispersed across paragraphs, tables, and experimental descriptions. In this survey, we identify four core tasks that constitute the data structurization pipeline. These tasks operate in a sequential manner, starting from entity identification and relation construction and culminating in the generation of structured outputs. Although each task has its own focus and research trajectory, together they enable the production of reliable, reusable data that supports downstream scientific analysis.

2.1.1. Named entity recognition

Named entity recognition (NER) is the first and most fundamental step of data structuring. It identifies, labels, and extracts predefined scientific entities from text, including material names, synthesis procedures, experimental parameters, and measured properties, which are

essential for materials modeling and data-driven research [16]. In addition to direct extraction, NER provides essential input to downstream tasks such as relation extraction, template filling, database construction, knowledge graph generation, research trend analysis, and automatic document classification.

Early NER systems primarily relied on rule-based methods, but recent studies have shifted to machine learning and transformer-based models. For example, Choudhary and Kelley [50] employed XLNet, a pre-trained transformer model, with the MatScholar dataset to extract material names, properties, and characterization methods. Similarly, Shetty et al. [20] used MaterialsBERT as an encoder-based model to extract material properties from polymer-related abstracts, achieving competitive performance. To further improve accuracy and generalization, researchers have explored more sophisticated architectures, such as integrating BERT embedding layers with BiLSTM networks. For example, Dai et al. [51] combined a BERT embedding layer, a BiLSTM layer, and a classification layer, paired with a GPT-assisted iterative training method, achieving an F1 score of 82.94% in materials science text. Other work, such as that by Wang et al. [52], leveraged part-of-speech tagging and relaxed dictionary matching in a BiLSTM-CRF model to extract alloy-related synthesis actions. More recently, researchers have examined whether LLMs can perform NER in few-shot or zero-shot settings. For example, Choi and Lee [53] used GPT-3 and GPT-3.5 to extract diverse material entities, and Kong et al. [54] applied GPT-3.5, GPT-4, and LLaMA-3 to extract material property measurements. As an illustrative example, Fig. 3 shows how an LLM-based NER system recognizes material property information (material, value, and unit) from a single sentence in the scientific literature. These studies collectively show that NER has evolved from rule-based solutions to more flexible transformer-based and LLM-based methods, enabling extraction at scale with reduced manual annotation.

2.1.2. Relation extraction

Since NER only identifies separate entities, an additional task is required to determine how they relate to each other. Relation extraction (RE) is therefore applied after NER to identify and classify semantic relationships among detected entities, such as those between a material and its properties or synthesis conditions [55]. For instance, in the sentence "$LiCoO_2$ batteries have an energy density of 150–200 Wh/kg," RE links the material entity $LiCoO_2$ to the property entity (energy density) and its value, as shown in Fig. 4. Together with NER, RE provides the structural basis for building databases, knowledge graphs, and other machine-readable scientific resources.

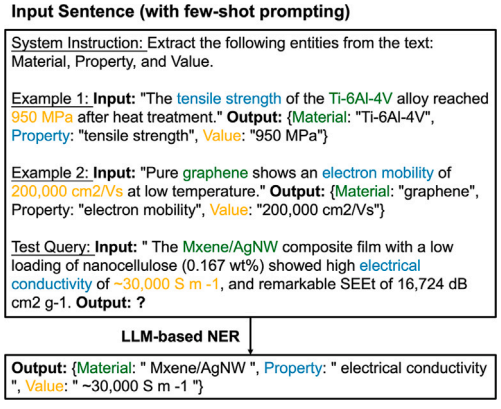


Fig. 3. Example of few-shot named entity recognition using an LLM.

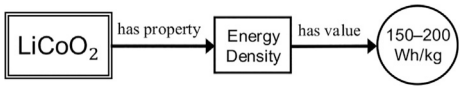


Fig. 4. Example of Extracted Relations.

Early domain-specific RE efforts included the adaptation of the ChemDataExtractor toolkit to battery materials [56]. Huang and Cole [57] modified the tool to generate more than 290,000 material–property–value records. Subsequent research investigated deep learning architectures. For instance, Shetty et al. [20] trained MaterialsBERT on polymer-related abstracts to extract material–property relationships. Mullick et al. [58] designed a pointer network–based encoder–decoder framework, which can extract multiple material–property triplets to populate a battery database. More recent work has shifted attention toward using LLMs as they support few-shot learning and are able to generalize across material domains. [9] fine-tuned GPT-3 and Llama-2 using prompting strategies to extract and classify material–property relationships in several contexts, including solid-state doping and metal–organic frameworks. These developments suggest that RE is moving from handcrafted systems to flexible and adaptive LLM-based methods.

2.1.3. Template filling

Once entities and relationships have been extracted, the next step is to transform this information into structured and reusable forms. Template filling is a downstream task of NER and RE that populates predefined template slots with extracted entities according to their relationships [55]. In materials science, this involves organizing material names, property values, units, and experimental parameters in a fixed format that can be used for database integration and automated scientific analysis, as shown in Fig. 1 Step 2. Specifically, Polak et al. [49] used LLMs from the GPT series with human assistance to transform the information related to the extracted materials into well-organized templates. [51] used GPT-3.5-turbo with prompting strategies to map unlabeled NER predictions into pre-designed dictionary-based templates. Template filling is particularly useful because it provides structured output that is consistent across studies, facilitating large-scale data aggregation and cross-publication comparisons.

2.1.4. Abstractive summarization

Unlike template filling, which focuses on predefined slot-based records such as materials, properties, and conditions, abstractive summarization in materials science is used to generate structured descriptive text fields. Although the output is textual, such summaries can still be regarded as structured information when they are generated in a standardized and reusable form, for example, as paper titles, contribution summaries, or brief descriptions of material systems and findings that support literature organization and decision-making. In this sense, abstractive summarization and template filling share a common goal: both transform unstructured scientific text into information units that can be stored, indexed, and used in downstream workflows, while differing mainly in the form of their outputs. Recent studies have explored this direction in materials science. For example, Choudhary and Kelley [50] applied a T5 model to generate publication titles from abstracts. Sayeed et al. [59] incorporated Gemini Pro as part of a JSON annotation workflow, enabling LLM-based abstractive summarization. Additionally, Bao et al. [60] designed a two-stage summarization framework using the Longformer model to produce comprehensive summaries. Despite these initial successes, abstractive summarization faces challenges due to limited domain-specific datasets and the need for expert validation to avoid factual hallucinations. Future progress will likely depend on task-specific summarization objectives, clearer definitions of target output fields, and evaluation protocols tailored to materials science use cases.

2.2. Multi-modal extraction

While the previous four tasks focus on structuring text-based information, a considerable portion of scientific knowledge in materials science publications is embedded in non-text formats such as equations, tables, figures, and microscopy images. Many key quantitative values, synthesis conditions, material structures, and performance indicators

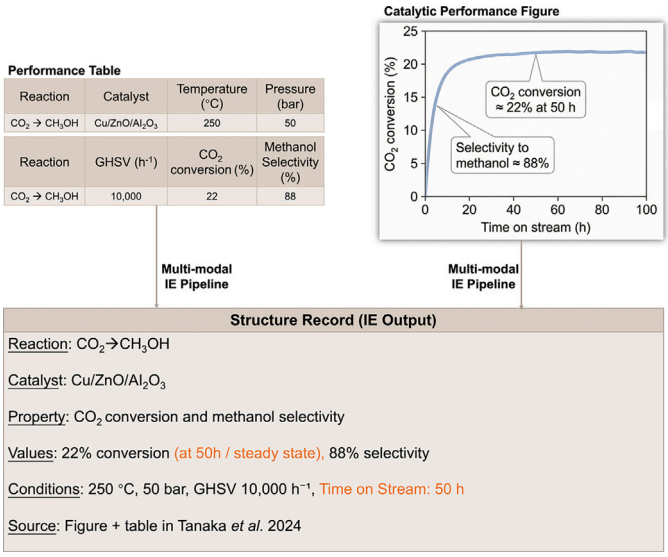


Fig. 5. Schematic example of multi-modal extraction in heterogeneous catalysis, where a catalytic performance plot and an accompanying table for CO_2 hydrogenation to methanol are converted into a unified structured record by an IE pipeline.

appear only in these modalities, making text-only extraction insufficient for capturing complete research insights. Fig. 5 illustrates a typical example of multi-modal IE in heterogeneous catalysis. A catalytic performance plot depicting time-on-stream behavior is interpreted alongside an accompanying table of experimental conditions. While the table provides precise numerical parameters (e.g., pressure and GHSV), the plot is essential for defining the temporal context of the reported performance, specifically confirming that the extracted values (22% conversion) represent a steady-state measurement taken at 50 hours, rather than transient initial data. Consequently, the final unified record captures both the operating conditions and the validated performance stability. This example highlights that many key quantities are only recoverable when figures and tables are interpreted together, rather than from text alone. Therefore, multi-modal extraction is necessary to extend IE beyond textual pipelines and enable comprehensive knowledge representation.

Despite its importance, multi-modal extraction in materials science remains relatively underexplored, and only a limited number of studies have investigated how to integrate multiple modalities into a unified IE framework. Recent research has begun to address this challenge. For example, Katzer et al. [61] combined transformer models with OCR, embedding models, an LLM, and a vision transformer to build a machine-readable database and chat-bot capable of ingesting multi-modal scientific information. In addition to such integrated pipelines, various works have targeted individual modalities. Tanishq Gupta et al. [62] fine-tuned the MatSciBERT model to extract material entities from image captions. Yi et al. [22] fine-tuned a GPT-based model for table extraction, transforming tables into machine-readable content and mitigating hallucinations through follow-up verification questions. Building on this direction, Polak and Morgan [63] proposed ChatExtract, a classification-based method that uses GPT-4 to extract structured data from tables and figure captions. Similarly, Kang et al. [64] introduced the LLM MOFs Miner, which employs a table agent to extract complex data from tables related to metal-organic frameworks (MOFs).

Although these studies show promising progress, current methods often treat modalities independently and lack unified frameworks capable of jointly reasoning over text, tables, and visual scientific evidence. For materials science, future work should emphasize tighter multi-modal fusion to jointly reason over text with domain-critical visual evidence, such as microstructure images, spectra, and phase diagrams, so that material properties and structures can be interpreted more consistently and reliably.

2.3. Classification

Beyond fine-grained entity extraction, some IE scenarios also require coarse-grained categorization to meet specific research needs. In this context, classification serves as a critical methodology to extract overarching themes or attributes by mapping documents, paragraphs, or sentences to predefined categories. Such high-level extraction is essential for the initial stages of materials discovery, supporting literature filtering, domain organization, and downstream automation pipelines such as trend analysis and database construction. At the document level, classification has been explored using both classical and transformer-based models. Choudhary and Kelley [50] employed statistical machine learning classifiers (e.g., SVM [65] and RF [66]) combined with a fine-tuned DistilBERT model to categorize materials science papers based on titles and abstracts. Bao et al. [60] developed a SciBERT-based classifier to organize dataset chapters into predefined categories. Large language models have also been tested for classification, with Polak and Morgan using GPT-3.5 and GPT-4 to determine whether sentences contained relevant numerical values for specific material properties [49].

Beyond document analysis, classification has also been applied at the paragraph level. Wang et al. [52] classified the paragraphs of the alloy synthesis procedure using a BERT-based model, while Tanishq Gupta et al. [62] used a BERT sentence classifier to assign abstracts in a materials dataset to specific research fields. These studies show that classification enables both knowledge filtering and large-scale literature organization, complementing structured IE tasks by improving data retrieval and navigation.

3. Methodology

This section reviews the methodological approaches used to perform information extraction (IE) tasks in materials science. Rather than categorizing methods solely by model architecture, we organize them based on how they address the practical requirements of materials-science IE. Specifically, we consider four complementary methodological paradigms. **Finetuning** enables models to incorporate domain-specific vocabulary and material knowledge through supervised adaptation to curated datasets. **Prompting** offers a lightweight alternative that leverages pretrained large language models without parameter updates, making it particularly suitable when labeled data are limited. **Retrieval-Augmented Generation (RAG)** addresses the limitations of static model knowledge by dynamically fetching relevant context from external corpora, thereby grounding generation in up-to-date literature. Finally, **Ensemble and agent-based methods** further enhance reliability by decomposing complex IE tasks into multi-step processes and enabling interaction with external tools or knowledge sources. Together, these approaches constitute the methodological foundation that makes LLM-based IE practical and effective for real-world materials research. Fig. 6 provides an overview of these four main patterns of LLM usage in materials-science IE: prompt-based inference, parameter tuning, retrieval-augmented grounding, and multi-component agent coordination.

3.1. Finetuning strategies

Many materials science tasks involve highly specialized terminology, complex physical concepts, or domain-specific relationships that general-purpose LLMs cannot fully capture. Finetuning addresses this limitation by updating a pretrained model using domain-specific training data [55]. This process allows the model to better understand the language of materials science, thereby facilitating more accurate data structuring and improving various downstream tasks, such as scientific question-answering (QA).

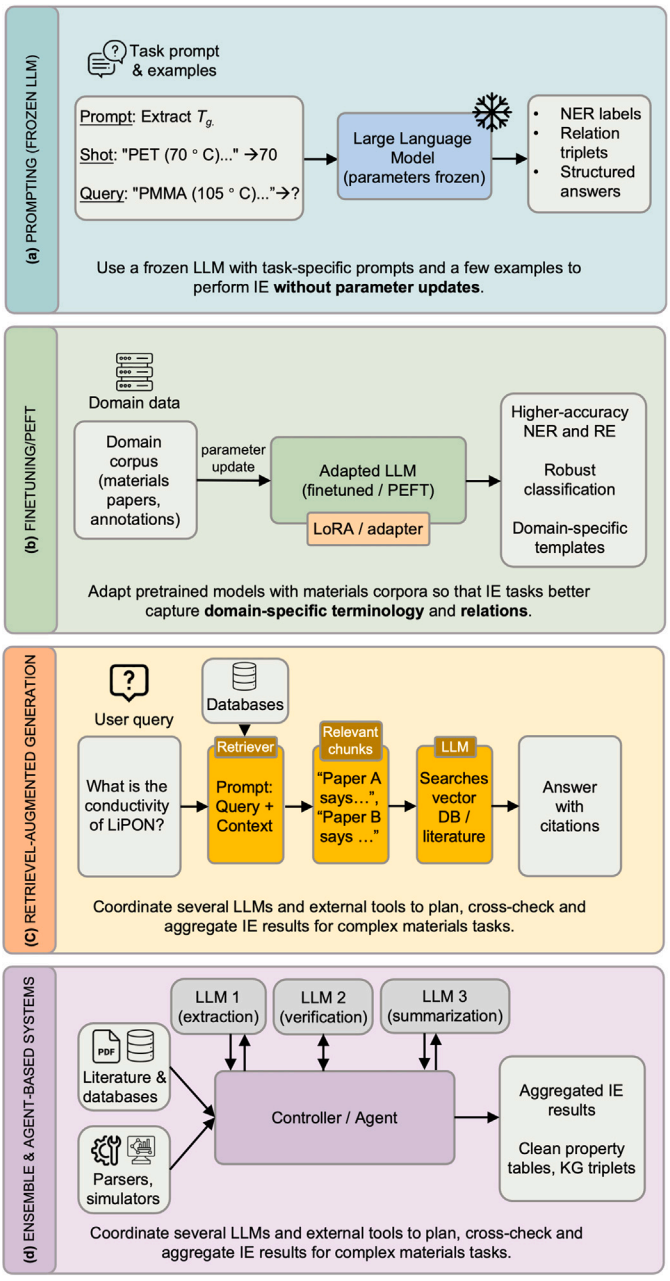


Fig. 6. LLM usage paradigms for materials-science information extraction. (a) **Prompting with frozen models** utilizes task-specific instructions without updates [54,67]. (b) **Finetuning/PEFT** adapts model weights using domain corpora to capture specialized terminology [9,53]. (c) **Retrieval-Augmented Generation (RAG)** dynamically retrieves external knowledge chunks to ground model outputs [68]. (d) **Ensemble and agent-based systems** coordinate multiple LLMs and tools [69].

Several studies provide concrete evidence that finetuning substantially improves model performance in materials science, particularly when adapting general-purpose language models to domain-specific terminology and reasoning patterns. Dagdelen et al. [9] fine-tuned Llama-2 and GPT-3 using only 400–650 annotated examples and achieved reliable extraction of material–property relations, such as impurity doping trends and metal–organic framework descriptors. Similarly, Choi and Lee [53] fine-tuned a GPT-3 model with domain instructions for battery and solid-state materials, improving its ability to identify relevant entities and conditions in the electrochemical literature. Beyond adapting general LLMs, domain-specific models further highlight the value of finetuning. Kumar et al. [71] fine-tuned MechBERT, a materials-oriented language model, on the SQuAD v2 dataset using a single question-answering head, enabling more accurate interpretation of mechanical stress–strain descriptions than general-purpose pretrained models. In summary, these results show that finetuning is an effective way to adapt pretrained models to the specialized language of materials science.

3.2. Prompting

In many materials science settings, large-scale labeled datasets required for finetuning are often unavailable. As a result, researchers are increasingly relying on **prompting**, which adapts LLMs to new tasks through task-specific instructions rather than parameter updates [55]. This paradigm is particularly well suited to scenarios where only one or two expert-annotated examples are available.

To meet materials-specific needs, several prompting strategies have been explored. Table 1 lists examples. Specifically, one useful strategy for adapting LLMs to materials tasks is **few-shot prompting**, where a small number of high-quality annotated examples are included in the prompt. These examples help the model interpret specialized materials terminology, such as mechanical descriptors, electrochemical parameters, or phase labels, that may not appear in its general training data. As a result, few-shot prompting is especially effective in domains where expert-labeled data are limited. Kong et al. [54] demonstrated this in a comparative study of material property NER under data-scarce conditions. They showed that GPT-4o achieved the highest accuracy when some curated examples were provided, outperforming both GPT-3.5 and conventional supervised models across multiple property types. Their results indicate that even a small number of high-quality examples can significantly improve the ability of LLMs to extract key material properties, such as elastic modulus, conductivity values, and phase-transition temperatures, directly from scientific text.

Beyond few-shot prompting, Prince et al. [70] used **zero-shot Chain-of-Thought prompting** with GPT-3.5 and Vicuna to construct a pipeline for experimental planning. Thway et al. [67] designed **question-based prompts** for GPT-3.5 to extract heating steps in solid-state synthesis into a predefined structured format. Building on this, Zhao et al. [68] applied **retrieval-augmented prompting**, which combines relevant text segments with the user query, allowing the model to produce more accurate and reliable outputs while reducing incorrect answers.

In general, prompting offers a practical and low-cost way to adapt pretrained LLMs to materials science IE tasks, reducing the need for extensive annotation while retaining strong performance on domain-specific extraction problems.

3.3. Retrieval-augmented generation

Retrieval-Augmented Generation (RAG) offers a distinct paradigm that grounds LLM responses in external, up-to-date knowledge. In a RAG workflow, the model retrieves relevant document chunks from a vector database based on the user's query and incorporates them into the context window before generating an answer. This approach is particularly critical for materials science QA and summarization tasks, where factual accuracy and citation provenance are critical.

While current LLMs are often criticized for producing incorrect, nonsensical, or fabricated responses (commonly referred to as hallucinations), RAG can help mitigate these issues by anchoring generative outputs to verified information fetched from external knowledge bases. For instance, Zhao et al. [68] developed LISA, a RAG-based system that retrieves information from a database of solid-state battery literature to answer complex queries about ionic conductivity and synthesis methods. By providing the LLM with specific paragraphs containing experimental evidence, LISA significantly reduced factual errors compared to standard prompting. Beyond QA, recent studies have expanded RAG to more complex workflows. For instance, Maharana et al. [77] developed a RAG pipeline to automate the construction of structured datasets like metal hydrides directly from scientific literature, achieving high accuracy

Table 1
Representative prompting strategies employed in materials science IE tasks, including few-shot prompting, zero-shot Chain-of-Thought prompting, question-based prompting, and retrieval-augmented prompting, with illustrative examples.

Strategy	Example Prompt Structure (Input)	Model Output / Rationale
Few-shot Prompting (e.g., Kong et al. [54])	*"Extract the yield strength.* *Example 1: 'The alloy showed 200 MPa yield stress.' → {Value: 200, Unit: MPa}* *Example 2: 'Yielding occurred at 1.5 GPa.' → {Value: 1.5, Unit: GPa}* ***Target:** 'The sample failed at 500 MPa.' → ?"*	*"{Value: 500, Unit: MPa}"* *(The model learns the JSON format from the provided examples.)*
Zero-shot Chain-of-Thought Prompting (e.g., Prince et al. [70])	*"Design an experiment to synthesize gold nanoparticles. **Let's think step by step** to ensure all precursors and safety constraints are met."*	*"**Step 1:** Select $HAuCl_4$ as the precursor…* ***Step 2:** Choose sodium citrate as the reducing agent…* ***Step 3:** Determine temperature…"*
Question-based (e.g., Thway et al. [67])	*"Extract the synthesis conditions from the text below.* *Q1: What is the base compound?* *Q2: What is the primary heating temperature?* *Text: … sintered at 900°C…"*	*"A1: Bi_2Te_3* *A2: 900°C"*
Retrieval-Augmented (e.g., Zhao et al. [68])	*"**Context:** [Retrieved snippet from literature regarding Li-ion conductivity in solid electrolytes…]* ***Question:** How does the annealing temperature affect ionic conductivity?"*	*"Based on the provided context, increasing the annealing temperature generally enhances ionic conductivity by promoting grain growth…"*

without manual curation. To rigorously evaluate such systems, Zhong et al. [78] introduced ChemRAG, a comprehensive benchmark for chemistry and materials QA, which revealed that retrieval-augmented models significantly outperform standard LLMs in handling domain-specific queries. Furthermore, Krotkov et al. [79] applied RAG to the design of nanostructured materials, showing that integrating laboratory protocols with scientific literature allows agents to generate feasible experimental designs. These frameworks establish RAG as a core methodological paradigm of LLM application in materials science (Fig. 6(c)), bridging the gap between static knowledge bases and dynamic generative capabilities.

3.4. Ensemble methods

Some materials science IE tasks remain difficult for a single model or a single prompting method. These challenges include synthesizing information across long documents, handling conflicting evidence, and extracting complex relationships such as linking host materials with dopants and mapping MOF structures to their properties. To address these domain problems, researchers have begun using **ensemble prompting** and **agent-based systems**. Analogous to expert consultation, these methods aggregate diverse reasoning trajectories and specialized knowledge to collectively determine the most accurate extraction result.

Ensemble prompting improves extraction stability by combining multiple reasoning styles, such as chain-of-thought, structured instructions, and example-based prompts, so that the strengths of one strategy can compensate for the weaknesses of another. Leiva-Araos et al. [80] used this approach with GPT-4-Turbo to summarize decades of materials literature, showing that an ensemble of prompts produced more coherent summaries and reduced missing contextual details compared to single-prompt baselines. This demonstrates that ensemble prompting can better capture nuanced scientific information in long, complex materials texts.

Agent-based systems address another type of materials problem: tasks that require planning, verification, or interaction with external tools. Instead of relying on a single model, these systems coordinate multiple agents that can call databases, perform consistency checks, or run external simulations. Ansari and Moosavi [69] developed an AI agent named Eunomia, which employed the ReAct architecture, GPT-4, and a Chain-of-Verification tool to extract difficult material–dopant relations and MOF descriptors with higher reliability. A related example comes from protein-based materials design, where complex property prediction and structure analysis require tight coupling between language models and external computational tools. Ghafarollahi and Buehler [81] introduced a multi-agent platform that integrates GPT-4 with physics simulators, generative models, and retrieval modules to design and analyze de novo protein materials.

Overall, ensemble and agent-based methods expand the capabilities of LLMs by addressing the challenges that materials science poses: complexity, ambiguity, long-range reasoning, and the need for cross-tool integration.

3.5. Comparative perspective

These methodological paradigms make different contributions to materials-science IE. Prompting is the most flexible and annotation-efficient approach, making it useful for rapid prototyping and low-resource extraction, but it is sensitive to prompt design and output formatting. Finetuning generally provides stronger performance for repeated tasks such as NER and relation extraction, but it requires curated annotations and may generalize poorly across distant materials subfields. RAG is particularly useful for literature QA, summarization, and provenance-aware extraction because it grounds outputs in retrieved evidence, although its performance depends strongly on retrieval quality. Ensemble and agent-based systems are promising for complex multi-step workflows, but they introduce higher computational cost, latency, and evaluation difficulty. Therefore, method selection should depend on the target IE task, available annotations, required reliability, and deployment constraints.

4. Datasets and resources

Progress in materials science IE is critically dependent on the availability and quality of resources that support both model development and evaluation. Compared with general NLP, which predominantly deals with generic textual data, materials literature presents distinct challenges: information is distributed across text, tables, figures, and simulation files; domain-specific terminology is highly specialized; and large-scale annotated datasets are scarce. These characteristics make well-designed resources indispensable for achieving accurate and scalable IE in materials science. Different types of resources play complementary roles in addressing these challenges. **Datasets** provide the empirical basis from which models learn to recognize materials entities, properties, and relations. Owing to the inherent heterogeneity of materials data, such datasets span a wide range of output formats and modalities, as schematically illustrated in Fig. 7. **Pretrained and foundation models** internalize foundational scientific knowledge and linguistic patterns through large-scale self-supervised learning on diverse corpora. This pre-acquired knowledge serves as a robust semantic prior, facilitating accurate extraction even under low-resource or limited-supervision settings. **Tools and frameworks** support components that LLMs cannot reliably handle in isolation, such as PDF parsing, table understanding, and multi-modal data processing. Finally, **evaluation protocols and benchmarks** ensure reproducibility and enable fair comparison among competing methods. Together, these resources

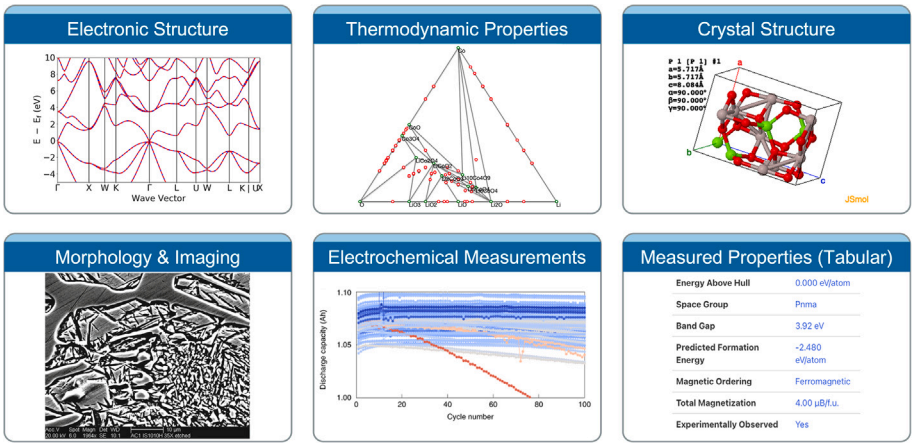


Fig. 7. Overview of six primary data modalities in materials science datasets. (a) **Electronic structure**: Band structure and DOS from JARVIS [72]. (b) **Thermodynamics**: Li–Co–O phase diagram from OQMD [73]. (c) **Crystal structure**: $MgAl_2O_4$ atomic model visualized from OQMD [73]. (d) **Morphology & Imaging**: Representative ultra-high carbon steel SEM image from UHCSDB [74]. (e) **Electrochemical measurements**: Charge–discharge curves from the Battery Life dataset [75]. (f) **Measured properties (Tabular)**: $LiFePO_4$ property records adapted from the Materials Project [76].

Table 2
Summary of datasets utilized in the surveyed materials science IE literature. Data types correspond to the modalities illustrated in Fig. 7.

Dataset	Source	Year	Subfield	Data Type	# Samples
AtomWork	Xu et al. [82]	2011	Inorganic materials	Crystal Structure	152,000
AtomWork Adv.	Xu et al. [83]	2024	Inorganic materials	Morphology & Imaging; Crystal Structure	1,705,011
AtomWork-Battery	Xu et al. [83]	2024	Battery materials	Electrochemical Measurements; Crystal Structure	40,763
Battery Materials	Huang and Cole [57]	2020	Battery material metadata	Measured Properties (Tabular)	292,313
CompES-X	Website [84]	2019	Inorganic compounds	Electronic Structure	55,839
CoRE MOFs 2019	Chung et al. [85]	2019	Porous MOFs	Crystal Structure; Measured Properties (Tabular)	14,000 +
OQMD	Kirklin et al. [73]	2015	Structural materials	Thermodynamic Properties; Crystal Structure	1,317,811
CSD	Groom et al. [86]	2016	Crystal structures	Crystal Structure	1,300,000 +

Table 3
Promising materials science databases for future IE research, categorized by the data types defined in Fig. 7.

Database	Source	Year	Subfield	Data Type	# Samples
AFLOW	Divilov et al. [87]	2025	Material compounds	Electronic Structure; Thermodynamic Properties	3,929,948
Battery Life	Severson et al. [75]	2019	Li-ion batteries	Electrochemical Measurements	124 (Cells)
JARVIS	Choudhary et al. [72]	2020	Multi-property materials	Electronic Structure; Crystal Structure	80,000 +
Materials Project	Jain et al. [76]	2013	Inorganic compounds	Tabulated Properties; Electronic Structure	154,000 +
NOMAD	Scheidgen et al. [88]	2023	DFT calculations	Electronic Structure; Crystal Structure	19,228,070
OCx24	Abed et al. [89]	2024	Catalysis	Tabulated Properties; Morphology & Imaging	20,000
Pauling File/MPDS	Villars et al. [90]	2019	Phase diagrams	Thermodynamic Properties; Crystal Structure	310,000
PolyInfo (NIMS)	Ishii et al. [91]	2024	Polymers	Measured Properties (Tabular)	552,427
UHCSDB	DeCost et al. [74]	2017	Steel microstructures	Morphology & Imaging (SEM Images)	961

Table 4
Datasets generated by studies reviewed in this survey, including task-specific resources for relation extraction, structured property collection, and domain-focused IE.

Database	Source	Year	Subfield	Data Type	# Samples
CCR	Polak et al. [49]	2024	Metallic glasses	Tabulated Properties	443
CO2RR	Choudhary et al. [72]	2023	CO_2 electroreduction	Tabulated Properties	3153
HEAs	Polak and Morgan [63]	2024	High-entropy alloys	Tabulated Properties	8900
MaTableGPT	Yi et al. [22]	2025	Electrocatalysis	Tabulated Properties	47,670
MatSciRE	Mullick et al. [58]	2024	Battery materials	Tabulated Properties (Relation Triplets)	11,000
MINDQUEST	Katzer et al. [61]	2025	Plasticity of FCC metals	Tabulated Properties	Not specified

constitute the practical foundation for building reliable IE systems in materials science.

4.1. Datasets

IE in materials science relies on structured datasets that capture materials, processing conditions, and properties across diverse subfields and data modalities. Tables 2–4 summarize three complementary categories of datasets that support current and emerging IE efforts. To clarify the heterogeneity of the information contained in these datasets, Fig. 7 presents a schematic overview of the primary output types they encompass. To bridge these conceptual categories with the surveyed resources, we explicitly map them to representative datasets listed in Tables 2–4. Specifically, these types include: (1) **Electronic structure** data, such as band gaps and density of states found in CompES-X [84]

and JARVIS [72]; (2) **Thermodynamic properties**, including phase diagrams and convex hulls central to the Pauling File [90] and OQMD [73]; (3) **Crystal structures**, represented by CIF-based geometric data [73, 86]; (4) **Morphology and imaging**, such as microstructural SEM images found in the UHCSDB dataset [74]; (5) **Electrochemical measurements**, particularly raw charge–discharge cycling curves characteristic of the Battery Life dataset [75]; and (6) **Tabulated material properties**, which form the core of structured repositories like The Materials Project [76] and PolyInfo [91]. This categorization highlights the breadth of information modalities that comprehensive IE systems must address.

The first group, shown in Table 2, consists of established materials databases that are directly used in the surveyed IE papers. These resources cover key domains such as inorganic crystal structures (AtomWork, CSD), battery performance and materials (AtomWork-Battery, Battery Materials), electronic structure and density functional theory (DFT) calculations (CompES-X, OQMD), and porous materials such as MOFs (CoRE MOFs 2019). Most of these datasets are organized as structured tables, often derived from experimental measurements or high-throughput calculations, and they provide the ground truth needed for tasks such as property extraction, template filling, and database population.

To illustrate the broader data landscape, Table 3 lists additional, well-established materials databases that are highly relevant for future multimodal IE research. Beyond standard structural repositories, this list includes resources that capture specific non-textual modalities: The Materials Project [76] and NOMAD [88] for comprehensive electronic structure and property records; UHCSDB [74] for microstructural SEM imaging; and the Battery Life dataset [75] for raw electrochemical charge–discharge cycling curves. Other complementary resources include PolyInfo for polymer structure–property data, the Pauling File / MPDS for thermodynamic phase diagrams, and OCx24 for catalysis data combining experimental and computational measurements. Collectively, these resources highlight the potential to build multimodal pipelines that link text not only with tables but also with images, raw experimental curves, and simulation outputs.

In addition to leveraging existing databases, several surveyed studies create new datasets as part of their IE pipelines, either through expert annotation, rule-based heuristics, or LLM-assisted labeling. These outputs are summarized in Table 4. They include corpora focused on named entity recognition and relation extraction, such as MatSciRE, structured numeric datasets targeting specific subfields such as high-entropy alloys and metallic glasses, and multimodal collections like MINDQUEST that integrate text, images, and tables. Although many of these datasets are relatively narrow in scope, they provide valuable task-specific benchmarks and training resources for materials IE. Together, the three groups of datasets, namely established materials databases, complementary large-scale resources, and datasets produced specifically for IE, define the current data landscape that both supports and constrains the development of information extraction methods in materials science.

4.2. Pretrained and foundation models relevant to materials-science IE

Pretrained language models play a central role in information extraction for materials science by transferring knowledge learned from large corpora to specialized extraction tasks. In practice, materials IE can build on both general-purpose language models and more specialized scientific or materials-domain models. Here, we focus on representative pretrained and foundation models that are most relevant to materials text understanding, structured extraction, and emerging multimodal reasoning.

Scientific and materials-text pretrained models. SciBERT provides a strong scientific baseline, as it is trained on a large scientific corpus and is better suited than general-domain language models for processing formula-like tokens, numerical expressions, and scientific writing patterns [34]. Beyond such broad scientific models, a growing number of text-oriented models have been pretrained specifically for materials science. MatBERT showed strong performance on materials NLP benchmarks and demonstrated the value of domain-adaptive pretraining on materials literature [92]. MaterialBERT was further developed for broad materials text processing [93], while MatSciBERT, trained on peer-reviewed materials science publications, outperformed SciBERT on named entity recognition, relation classification, and abstract classification benchmarks in the materials domain [62].

A growing number of models have also been developed for narrower materials subdomains. MaterialsBERT, trained on 2.4 million materials science abstracts and used in a polymer property extraction pipeline, improved performance on several NER benchmarks and demonstrated the usefulness of continued pretraining for targeted extraction tasks [20]. BatteryBERT showed that battery-specific pretraining improves downstream tasks such as battery paper classification, extractive question answering, and battery database enhancement [94]. MechBERT further illustrates the value of subdomain adaptation for mechanics-oriented text, where a model specialized for mechanical stress–strain language improves performance on tasks requiring understanding of deformation descriptions in materials literature [71].

Materials-specific foundation models. Recent work has expanded from text-oriented pretrained models toward broader materials foundation models. LLaMat is developed through continued pretraining of LLaMA models on a large corpus of materials literature and crystallographic data, and it is evaluated on materials-specific NLP and structured information extraction tasks [95]. MatterGPT is a GPT-style generative transformer trained on SLICES representations for inverse design of solid-state materials. Although its primary goal is generation rather than IE, it represents an important shift toward materials-native foundation models [96]. MatterChat extends this direction further by combining textual inputs with structure-aware materials information in a multimodal LLM framework [97]. These broader foundation models are not all designed specifically for information extraction, but they are increasingly relevant to future IE systems that must integrate text with structural, tabular, and other non-textual materials data.

Overall, pretrained and foundation models form the basis of modern IE pipelines in materials science. Scientific and materials-text pretrained models provide the strongest support for current text-based IE tasks, while broader materials foundation models suggest that future IE systems will increasingly move beyond text-only extraction toward more integrated, structure-aware, and multimodal forms of knowledge acquisition.

4.3. Tools and frameworks

To effectively apply information extraction methods in materials science, researchers rely on a range of external tools that address the limitations of text-only language models. These tools support key functions such as document preprocessing, multimodal data extraction, and structured data generation, and they often serve as integral components of agent-based or hybrid information extraction pipelines. From a functional perspective, such tools can be broadly organized into three categories, namely text preprocessing and PDF parsing, table and figure extraction, and frameworks that support multimodal processing and agent-based workflows.

(a) Text preprocessing and PDF parsing. Large language models require clean and machine-readable text as input, whereas scientific articles are typically distributed in PDF format. Tools such as [98] address this mismatch by converting PDF documents into structured XML or TEI representations, which enable reliable downstream information extraction tasks including named entity recognition, relation extraction, and structured template filling. This capability is particularly important in materials science, where experimental conditions, synthesis procedures,

and property descriptions are often embedded within complex multi-column layouts that cannot be processed accurately without dedicated preprocessing.

(b) Table and figure extraction. A substantial portion of materials science data, including performance metrics, electronic structure calculations, and structural parameters, is reported in tables or embedded within figures rather than in the main text. General-purpose table extraction tools such as Camelot Mehta et al. [99] and Tabula [100] convert tabular regions in PDF documents into structured formats such as CSV or JSON that can be directly integrated into information extraction pipelines. For visual content, tools such as DeepFigures [101] and Donut [102] enable figure detection, optical character recognition, and diagram-to-text conversion. These capabilities are essential for materials science, where key experimental evidence, including diffraction patterns, microscopy images, and performance curves, is frequently conveyed through figures rather than textual descriptions.

(c) Multi-modal and agent-supporting frameworks. Recent multimodal IE systems increasingly integrate external tools with large language models through orchestration frameworks and agent-based designs. For example, Katzer et al. [61] combined OCR tools such as Marker, the all-MiniLM embedding model, and the LLaVA vision transformer with a large language model to construct a machine-readable database of multiple sources from text, tables, and figures. This example illustrates how external components complement large language models by providing capabilities that remain difficult for text-centric models, including the ability to extract information from plots and interpret crystallographic diagrams. In agent-based pipelines, these tools are typically invoked programmatically through modular steps. An agent can call a parser to isolate and extract table regions, another can analyze diffraction peak positions using domain-specific software, and the language model can then integrate the intermediate output into consistent structured representations.

In general, these tools constitute the backbone of practical information extraction workflows in materials science. Although large language models provide strong semantic understanding, they continue to rely on specialized external utilities to process complex document formats, extract information from multimodal sources, and support domain-specific extraction tasks. The integration of such tools into unified frameworks, either through structured pipelines or agent-based systems, therefore represents a critical direction for the development of scalable, robust, and reliable information extraction systems for materials science.

4.4. Evaluation & benchmarks

Reliable evaluation is essential for developing IE systems that can be trusted in materials science research. Since materials literature contains highly specialized terminology and multi-modal data, achieving accurate, context-dependent measurements is inherently difficult; consequently, standard NLP metrics are often insufficient unless they are specifically adapted to the materials domain. Therefore, materials IE evaluation must consider not only accuracy, but also domain relevance, factual correctness, and generalization between different classes of materials.

For sequence labeling tasks such as material, property, or process NER, F1-score remains the primary metric. It provides a balanced measure of Precision and Recall, which is critical for the class-imbalanced and high-specificity terminology found in materials science. Benchmarks such as MatRecData explicitly adopt the F1-score to evaluate the quality of extraction at the entity level [103]. These metrics are particularly important because missing or mislabeling an entity (e.g., a dopant, phase label, or synthesis parameter) can significantly distort downstream analyses, such as trend identification or database construction.

Relation extraction aims to link materials, properties, and experimental conditions into structured triplets. Evaluation is commonly conducted using triplet-level Precision, Recall, and F1 scores, where a predicted relation is considered correct only when all constituent elements exactly match the ground truth. The MatSciRE benchmark adopts this evaluation protocol for relationships such as material conductivity pairs and battery capacity triplets, thus providing a rigorous and structured framework for assessing relation extraction models [58]. This form of evaluation is essential in materials science, since even minor discrepancies, such as inconsistent units or incorrect contextual associations, can fundamentally alter the scientific interpretation of an extracted relation.

Generative tasks such as abstractive summarization and structured template filling require evaluation strategies that differ from those used for extraction tasks because their outputs are inherently free-form. Automated metrics including ROUGE and BERTScore are commonly employed to measure content overlap and semantic similarity between generated outputs and reference texts. However, in the context of materials science, these metrics alone are insufficient and must be complemented by human evaluation to ensure factual correctness and scientific validity. For instance, consider a scenario where a model generates “The enhanced stability is attributed to **Ni**-doping” instead of the ground truth “**Ti**-doping.” Standard metrics like ROUGE would assign a high score due to the near-identical lexical structure, and BERTScore might indicate high similarity as both are transition metals; yet, this constitutes a factual error that fundamentally misrepresents the material’s chemistry.

For example, Park et al. [104] conducted expert-based assessments of the summaries generated by GPT-3.5 using criteria that emphasized technical precision, content relevance, and structural clarity. Such human-in-the-loop evaluation has become increasingly important, as large language models are prone to producing fluent yet scientifically inaccurate statements.

Although several datasets, including MatRecData [103] and MatSciRE [58], provide valuable resources for evaluation, the field still lacks comprehensive and community-accepted benchmarks that span multiple subdomains, data modalities, and task categories. Existing benchmarks for materials information extraction are typically confined to narrow problem settings, such as battery-related properties or polymer abstracts, and offer limited interoperability across datasets and tasks. As a consequence, it remains challenging to compare models in a fair and systematic manner or to assess progress across the field in a consistent way. This absence of standardized benchmarks represents a major obstacle to reproducible research and underscores the need for broader and more unified evaluation suites in future work.

Overall, evaluation in materials information extraction requires a carefully balanced combination of quantitative metrics, expert-driven assessment, and robustness analysis across diverse domains. The development of richer, multimodal, and community-endorsed benchmarks therefore remains a critical opportunity for advancing reliable and deployable large language model applications in materials science.

5. Applications

Information extraction underpins several important downstream applications in materials science by transforming unstructured scientific literature into structured and machine-interpretable knowledge. This section focuses on three representative application scenarios, namely database construction, material trend analysis, and knowledge graph generation, through which extracted entities and relations can be systematically exploited to support data-driven materials discovery. Curated databases and knowledge graphs rely primarily on named entity recognition, relation extraction, and template filling, while trend analysis also depends on document-level classification to identify, filter, and aggregate relevant records across large document collections.

It should be noted that current LLM-based IE studies are unevenly distributed across materials subfields. Batteries, polymers, alloys, and catalysis are more frequently represented in the existing literature because they have relatively well-defined property targets, established

databases, and strong demand for large-scale literature curation. In contrast, ceramics, biomaterials, and semiconductors remain underrepresented in current IE studies, although they involve important extraction targets such as sintering conditions, phase composition, biocompatibility, degradation behavior, band gaps, defects, doping, and carrier transport properties. This imbalance reflects the current availability of datasets and benchmarks rather than the intended scope of materials IE, and expanding coverage to these underrepresented areas is an important direction for future research.

5.1. Database construction

Database construction is a key application of information extraction, where entities and relations from NER and RE tasks are systematically stored and organized into structured databases. The availability of these databases is foundational to modern materials science, empowering researchers to perform data-driven research, including materials design, property prediction, and knowledge discovery. Early efforts, such as those by Huang and Cole, used ChemDataExtractor to perform NER and RE for a battery dataset with over 290,000 entries [57]. They utilized this structured data to reveal historical trends in battery research, quantitatively mapping the shift in focus from lead-acid to lithium-ion chemistries over the past century. Building on this, deep learning models emerged; Mullick et al. trained a pointer-network model with MatBERT embeddings to extract entity–relation triples [58], while Choudhary and Kelley fine-tuned an XLNet model to create a database focused on inorganic materials [50], which facilitated the prediction of synthesis parameters and bandgaps for novel solid-state compounds.

More recent methods have integrated LLMs and agent-based systems. [83] developed the Template Oriented Atomic Simulation Toolkit, which consolidates data on chemical compositions, crystal structures, and battery performance into a comprehensive database, enabling the screening of high-performance electrolytes based on aggregated ionic conductivity values. Similarly, [64] applied the L2M3 model, an agent-based system, to extract information from 40,000 papers on metal–organic frameworks and integrate it with the Cambridge Structural Database. This integration allowed for the identification of structure-property relationships governing gas adsorption capacities in porous materials. [63] introduced ChatExtract, which uses CoT prompting to construct experimental databases for metallic glasses and high-entropy alloys. Using these extracted records, they successfully predicted critical cooling rates and yield strengths, demonstrating that LLM-curated data can rival human-curated datasets for training predictive machine learning models. These efforts provide a bridge between raw text and structured knowledge, accelerating discovery and supporting predictive modeling.

5.2. Material trend analysis

Material trend analysis examines how the prominence of materials and related research topics evolves over time within the scientific literature. This application is enabled by NER and subsequent data structuring, and allows researchers to identify emerging materials, anticipate changes in research directions, and prioritize research resources more effectively.

Several studies have demonstrated the effectiveness of this entity-centric approach. For example, Polak and Morgan [63] fine-tuned GPT-3.5-turbo using few-shot prompting to perform named entity recognition and structured template filling, which enabled the analysis of yearly trends in material mentions and material structures over the period from 2014 to 2023. Building on this idea, Choi et al. [105] applied MatBERT to NER on publications related to electrochemical carbon dioxide reduction, thereby supporting longitudinal analyses of catalyst design strategies and elemental composition patterns from 1997 to 2021. Similarly, Shetty et al. [20] fine-tuned MaterialsBERT to extract entities from polymer science literature, revealing temporal trends in polymers associated with applications such as solar cells and fuel cells. Their study further introduced domain-specific visualizations that captured the evolution of performance trade-offs, including analyses of energy and power density characteristics in supercapacitor research.

Together, these applications illustrate how information extraction enables a data-driven view of evolving research priorities and provides actionable guidance for future directions in materials discovery.

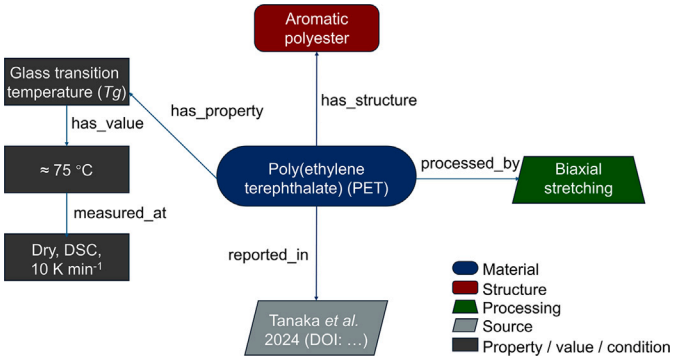


Fig. 8. Example slice of a polymer knowledge graph populated from IE outputs, showing how a structural polymer (PET) is linked to its family, glass-transition property, processing conditions, and literature source through typed relations.

5.3. Knowledge graph construction

A knowledge graph provides a structured representation of information by organizing entities as nodes and their relationships as edges, enabling systematic storage, retrieval, and reasoning over complex knowledge. In materials science, knowledge graphs are typically constructed by leveraging the outputs of NER and relation extraction to represent material-centric relationships, integrate heterogeneous data sources, and ultimately support accelerated materials discovery. Fig. 8 presents a schematic view of a polymer knowledge graph derived from IE outputs. For poly(ethylene terephthalate) (PET), the graph connects the material to its polymer family, glass transition temperature, measurement conditions, processing route, and source literature through explicitly defined relations. Such representations provide a compact and interpretable view of structure, property, and processing relationships that can be queried and analyzed at scale.

Several studies have demonstrated the effectiveness of this approach. Liu et al. [28] proposed ASaRE-Net, which integrates a pretrained RoBERTa model to jointly perform named entity recognition and relation extraction in Chinese journal articles related to aluminum silicon alloys, and uses the extracted triples to construct a materials knowledge graph. This graph enabled researchers to visually analyze the dependencies between heat treatment parameters and mechanical properties (e.g., tensile strength), facilitating the identification of optimal processing routes for specific alloy grades. Similarly, Dai et al. [51] employed a fine-tuned GPT-3.5-turbo model together with a Neo4j database to build a knowledge graph, illustrating how LLMs can directly support structured knowledge integration. By performing multi-hop reasoning over the graph, their system could infer implicit connections between synthesis methods and resulting microstructures, revealing insights that were not explicitly stated in any single document. In another line of work, Dreger et al. [106] developed a pipeline that combines a GPT-based model with rule-based scripts to extract relations from tabular catalyst data and transform them into a knowledge graph. This structured representation allowed for the systematic identification of performance trends across different catalyst compositions, helping to pinpoint under-explored chemical spaces for reaction optimization.

Collectively, these studies show that knowledge graphs constitute a powerful application of IE, enabling the transformation of dispersed textual and tabular information into interconnected knowledge structures

that support advanced querying, analysis, and reasoning in materials research.

6. Challenges and opportunities

Despite the rapid progress outlined in the preceding sections, substantial challenges remain before LLM-based information extraction can be reliably deployed across the materials science domain. This section highlights six key challenges and future priorities that currently constrain the effectiveness of LLM-based information extraction for materials science. These challenges include the limited availability of high-quality annotated datasets, the absence of standardized evaluation protocols tailored to materials-specific tasks, the inherent complexity and inconsistency of materials terminology, uneven coverage across materials subfields, the difficulty of integrating information from non-textual sources such as figures, spectra, and tables, and practical deployment issues such as cost, latency, API dependency, privacy, and model version drift. Addressing these issues is critical for developing more accurate, comprehensive, and reliable extraction systems and for enabling the next generation of data-driven materials discovery.

6.1. Inconsistent material terminology

Materials science literature exhibits extremely diverse and inconsistent terminology. The same compound may appear under multiple names, abbreviations, or chemical formulas (e.g., "$LiCoO_2$", "LCO", or "lithium cobalt oxide"), with doping and composite materials introducing further ambiguity. This inconsistency severely impacts NER and relation extraction, ultimately hindering knowledge graph construction, as LLMs must accurately map these heterogeneous expressions into canonical forms. Supporting this claim, Kong et al. [54] noted that chemical entity extraction is particularly difficult when training labels are generated automatically, as current automated tagging methods often fail to resolve linguistic complexity. A potential solution involves developing comprehensive entity dictionaries to interpret arbitrary abbreviations and resolve complex nomenclature, a strategy advocated by Yi et al. [22]. However, materials science currently suffers from a scarcity of such comprehensive ontologies and synonym dictionaries. Therefore, developing standardized vocabularies, robust ontologies, and automatic normalization tools is crucial to improving the reliability and interpretability of extraction results.

6.2. Uneven domain coverage across materials subfields

The studies reviewed in this survey suggest that LLM-based IE is not yet evenly distributed across materials science. Batteries, polymers, alloys, and catalysis have received more attention because they are associated with clearer property targets, richer structured databases, and stronger demand for large-scale literature curation. However, other important subfields, including ceramics, biomaterials, and semiconductors, remain less represented. This limited coverage may reduce the generalizability of existing IE methods, since these areas involve different terminology, data formats, and extraction targets. For example, ceramics often require the extraction of sintering conditions, grain size, phase composition, and thermal or mechanical properties; biomaterials require information on biocompatibility, degradation, scaffold morphology, and biological response; and semiconductors require extraction of band gaps, defect chemistry, doping conditions, carrier mobility, and device-relevant properties. Future materials IE systems should therefore be evaluated across a broader range of material classes, rather than relying only on the subfields where current datasets are most available.

6.3. Integration with non-textual modalities

A major challenge for current LLMs in materials informatics lies in the effective integration of non-textual modalities. A substantial fraction of scientifically important information is not explicitly described in the main text, but is instead encoded in figures, tables, and spectroscopic data. Crucial quantitative evidence is frequently embedded in graphical forms, including diffraction patterns from which phase composition and crystallinity are inferred, microscopy images that convey morphology and crystal structure, and spectroscopic measurements that reveal chemical states and bonding environments. In addition, performance curves like stress–strain responses and battery charge–discharge profiles often provide the primary evidence of a material's functional value. Despite these requirements, most current LLMs for materials informatics remain inherently text-centric. This architectural focus limits their ability to natively recognize and interpret diverse data representations across different modalities. For example, Thway et al. [67] mentioned the limitation of using GPT-3.5 alone to extract material analysis information from diffraction images, which requires specialized tools for automated plot reading [107] and refinement [108]. Therefore, a critical future direction involves developing multimodal IE methods that effectively combine text, table, and image representations to enable a more comprehensive materials understanding, thereby enhancing database creation and accelerating materials design automation.

6.4. Data scarcity

To automatically resolve terminological inconsistencies and integrate non-textual modalities, a promising approach is to tune LLMs on expansive corpora, which can cultivate a "human-like" understanding of complex material relationships through self-supervised learning. However, achieving such robust semantic alignment requires an immense volume of high-quality data; for instance, in the general NLP domain, GPT-3 [109] was trained on 570 GB of text, Llama-3 [110] leveraged over 15 trillion tokens. In contrast to the massive unlabeled corpora used in general NLP, task-specific materials science datasets are typically small in scale, heterogeneous in structure, and costly to construct, creating a fundamental barrier to the development of reliable domain-specific applications. For example, gold-standard annotated corpora often comprise only a few hundred documents (e.g., < 5 MB of text), as accurate annotation requires substantial domain expertise [111,112]. This data scarcity restricts supervised training and domain specific fine-tuning, which in turn hampers model generalization across diverse classes of materials, such as transitions from organic polymers to inorganic alloys. For example, Jalali et al. [111] showed that fine-tuning GPT on an electronic laboratory notebook resulted in limited performance gains because the available annotations did not provide sufficient coverage of the relevant domains of materials. Addressing this challenge will require sustained efforts to build open and large-scale annotated corpora, as well as the development of weakly supervised and self-supervised learning strategies that can reduce dependence on exhaustive manual annotation.

6.5. Lack of MatSci-specific evaluation protocols

Despite rapid progress in the application of large language models to materials science, the field continues to lack standardized evaluation protocols. Benchmarks developed for general natural language processing are insufficient for assessing domain-specific tasks in materials science, such as the identification of synthesis parameters or the extraction of property and composition relationships. This deficiency arises because material data are inherently context-dependent, with factors such as temperature and pressure playing a critical role in determining meaning and validity, which makes the definition of reliable "ground truth" particularly challenging. The absence of unified and context-aware evaluation benchmarks therefore limits reproducibility and complicates fair comparisons between models [59]. Addressing this limitation will require coordinated community efforts to develop evaluation datasets and metrics that more accurately reflect the scientific complexity of the literature on materials.

6.6. Practical deployment issues of LLM-based IE

Beyond task-level accuracy, practical deployment remains a key barrier for LLM-based IE in materials science. Large proprietary models can introduce substantial computational costs and latency when applied to large literature collections, especially when multi-step prompting, retrieval-augmented generation, or agent-based verification is used. Similar concerns have been widely discussed in the broader foundation-model literature, where large models raise issues related to computational resources, environmental and financial costs, and deployment risks [113,114]. These constraints may limit large-scale corpus processing and make results difficult to reproduce across laboratories with different computational resources.

API-dependent systems introduce additional concerns. Model updates can change outputs over time, creating version drift and making published IE pipelines difficult to reproduce; this issue has been demonstrated for GPT-3.5 and GPT-4 services, whose behavior and performance can change substantially over time [115]. Privacy is also important when models are applied to unpublished manuscripts, proprietary industrial reports, or internal laboratory notebooks, as recent surveys have highlighted privacy risks in LLM and LLM-agent systems [116]. Future systems should therefore report model versions, prompts, retrieval settings, and cost-related information, and should consider local or open-weight models when privacy, reproducibility, or long-term stability is required.

6.7. Prioritized roadmap for future materials IE

In the short term, the field should focus on terminology normalization, clearer extraction schemas, and expert-validated benchmarks. These foundations are necessary for reliable comparison across models and materials subfields, especially when extracted records contain material names, processing conditions, numerical properties, and evidence spans.

In the medium term, future work should develop multimodal IE systems that jointly process text, tables, spectra, microscopy images, and performance curves. This is essential because many material properties and experimental conditions can only be interpreted by combining multiple evidence sources, such as linking synthesis text with tabulated conditions or associating performance values with curves and figure captions.

In the long term, materials IE should move toward provenance-aware and tool-integrated systems that combine LLMs with databases, parsers, simulation tools, and human validation. Such systems should not only extract information, but also verify values, track evidence sources, normalize units, and support reliable database construction, trend analysis, and knowledge graph generation.

7. Conclusion

This survey reviews the rapid integration of LLMs into materials science information extraction and highlights their role in transforming unstructured literature into structured and reusable knowledge. We summarize key IE tasks, including named entity recognition, relation extraction, template filling, abstractive summarization, multi-modal extraction, and classification, and review major methodological paradigms such as prompting, finetuning, retrieval-augmented generation, and ensemble or agent-based systems. We also discuss datasets, pretrained and foundation models, tools, evaluation protocols, and representative applications including database construction, material trend analysis, and knowledge graph generation.

Despite this progress, several challenges remain before LLM-based IE can be reliably adopted across the full materials science domain. These include inconsistent material terminology, uneven coverage across materials subfields, limited integration with non-textual modalities, scarcity of high-quality annotated data, insufficient materials-specific evaluation protocols, and practical deployment issues such as cost, latency, API dependency, privacy, and model version drift. Addressing these challenges will require clearer extraction schemas, broader and expert-validated benchmarks, stronger multimodal pipelines, provenance-aware evaluation, and tool-integrated systems that combine LLMs with materials databases and domain-specific analysis tools.

Overall, LLM-based IE provides a promising foundation for scalable literature understanding and data-driven materials discovery. Future progress will depend not only on more capable models, but also on reliable workflows that can extract, verify, and organize materials knowledge in forms that are useful for scientific analysis and decision-making.

CRediT authorship contribution statement

Yutong Duan: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ying Zhang:** Writing – review & editing, Supervision, Conceptualization. **Runjia Yu:** Investigation. **Dongyuan Li:** Writing – review & editing, Supervision, Conceptualization. **Satoshi Kosugi:** Supervision. **Manabu Okumura:** Supervision. **Kotaro Funakoshi:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data were used for the research described in the article.

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